25900774 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCCHE100 - FOOD CHEMISTRY AND NUTRITION

(2024 ADMISION ONWARDS)

Duration: 0.75 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is a trace mineral? [K] / [CO1]

   a). Calcium b). sodium

   c). potassium d). Zinc

2. Folic acid is also known as: [K] / [CO1]

   a). Vitamin B9 b). Vitamin B6

   c). Vitamin A d). Vitamin C

3. Which protein is most abundant in the human body? [U] / [CO1]

   a). Myosin b). Collagen

c). Keratin d). Hemoglobin

4. What is the function of food? [U] / [CO1]

a). Nutrition b). Energy

c). Hydration d). All options are correct

5. Identify the substance which is NOT used as a gelling agent from the following. [K] / [CO2]

a). Gelatin b). Pectin

c). Agar agar d). Chlorophyll

6. Which type of fat should be limited in the diet due to its negative effect on heart health? [U] / [CO1]

a). Polyunsaturated fats b). Monounsaturated fats

c). Saturated fats d). Trans fats

7. Identify the thickener used in soups and sauces. [K] / [CO2]

a). Corn starch b). Tannin

c). Shellac d). Citric acid

8. Which among the following can be used as a stabilizer in meat products? [K] / [CO2]

a). Carrageenan b). China grass

c). Acetic acid d). Salt

9. Why are E numbers used on food labels? [K] / [CO2]

a). To indicate that the food is organic b). To simplify the identification of food additives

c). To warn of foodborne illnesses d). To classify foods as processed

10. Which food additive has been used by ancestors for preserving food? [U] / [CO2]

a). Sodium Chloride b). Sodium Benzoate

c). Sodium Nitrate d). Ethanoic Acid

11. Which natural substance, often found in egg yolk, acts as an emulsifier? [U] / [CO2]

a). Pectin b). Lecithin

c). Gelatin d). Casein

12. Identify the substance which is NOT used as a flavour enhancer. [K] / [CO2]

a). Ajinomoto b). Salt

c). Citric acid d). Sodium benzoate

13. Name the additive which provides a firm but elastic consistency to food. [U] / [CO2]

a). Gelling agents b). Glazing agents

c). Curing agents d). Coloring agents

14. Select the non-caloric sweetner among the following. [K] / [CO2]

a). Fructose b). Sucrose

c). Aspartame d). Glucose

15. Select the statement which is NOT TRUE about food preservatives. [U] / [CO2]

a). They prevent food spoilage b). They prevent microbial growth

c). They provide nutritional value d). They prevent undesirable changes

16. Identify the food color which is obtained from carrot. [K] / [CO2]

a). Carotenoids b). Carmine yellow

c). Brilliant blue d). Sudan red

17. Identify an emulsifier which is used in sauces. [K] / [CO2]

a). Xanthan Gum b). Sorbic acid

c). Mannitol d). Carotenoids

18. Which carbohydrate is commonly known as table sugar? [K] / [CO1]

a). Lactose b). Maltose

c). Glucose d). Sucrose

19. What health risk is associated with consuming turmeric adulterated with lead chromate? [U] / [CO3]

a). Liver damage b). Bone fractures

c). High blood pressure d). Skin infections

20. What other common adulterant may be added to black pepper to increase its weight? [U] / [CO3]

a). Black stones b). Chalk powder

c). Black dye d). Dirt particles

21. Which of the following is an example of an unintentional adulterant? [U] / [CO3]

a). Sand b). Water

c). Dyes d). Heavy metals

22. Which of the following substances is commonly added adultrant in chili powder but is unsafe for consumption? [K] / [CO3]

a). Red clay powder b). Papaya seeds

c). Sawdust d). Crushed red pepper

23. Artificial color in tea can lead to which of the following health issues? [U] / [CO3]

a). Stomach discomfort b). Vision problems

c). Hearing loss d). Heart palpitations

24. Which health problem is associated with the consumption of butter adulterated with hydrogenated fats? [K] / [CO3]

a). Digestive disorders b). High cholesterol

c). Fever d). Vision issues

25. Vanaspati is a common adulterant in ghee. Which health issue is commonly associated with its consumption? [U] / [CO3]

a). Stomach cramps b). High cholesterol

c). Skin rash d). Fever

26. Why is chalk powder often added as an adulterant in sugar? [U] / [CO3]

a). To reduce weight
b). To add sweetness
c). To make it appear whiter and bulkier
d). To add minerals

27. How can you detect the presence of water in milk as an adulterant? [K] / [CO3]

a). Taste the milk
b). Freeze a small quantity of milk
c). Boil the milk and observe the color change
d). Measure the density or use a lactometer

28. Which health effect is commonly associated with the consumption of food adulterated with lead compounds? [K] / [CO3]

a). Cancer
b). Kidney failure
c). Brain damage, especially in children
d). Dehydration

29. What is the significance of food labeling under the Food Adulteration Act? [U] / [CO4]

a). It attracts more customers
b). It provides nutritional information
c). It ensures transparency and safety
d). It increases the shelf life of food

30. What type of food is typically classified as "fast food"? [U] / [CO4]

a). Fresh salads
b). Whole grain sandwiches
c). Pre-prepared and quickly served meals
d). Home-cooked meals

31. Traditional food habits are often considered healthier because they. [U] / [CO4]

a). Use fewer spices and flavors
b). Emphasize fast preparation and convenience
c). Involve more whole, unprocessed ingredients
d). Rely on artificial additives for taste

32. Which ingredient in soft drinks is primarily responsible for their high sugar content? [U] / [CO4]

a). Sucrose
b). Fructose

c). High-fructose corn syrup d). All options are correct

33. What is a common psychological effect of consuming junk food? [U] / [CO4]

a). Increased stress b). Improved mood

c). Enhanced focus d). Better sleep

34. Which health risk can arise from consuming wheat adulterated with stone particles? [U] / [CO3]

a). Intestinal blockage b). Liver toxicity

c). Increased sugar levels d). Loss of appetite

35. Which technique is commonly used to detect adulteration of honey with invert sugar syrup? [K] / [CO3]

a). Benedict's solution test b). Thin-layer chromatography

c). Specific gravity test d). Furfural test

[35X1=35]

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